

CSE 676 – Deep Learning

Assignment 2 Report

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1 R.1 Training Curve

Figure 1 shows the training and validation loss over 5 epochs.

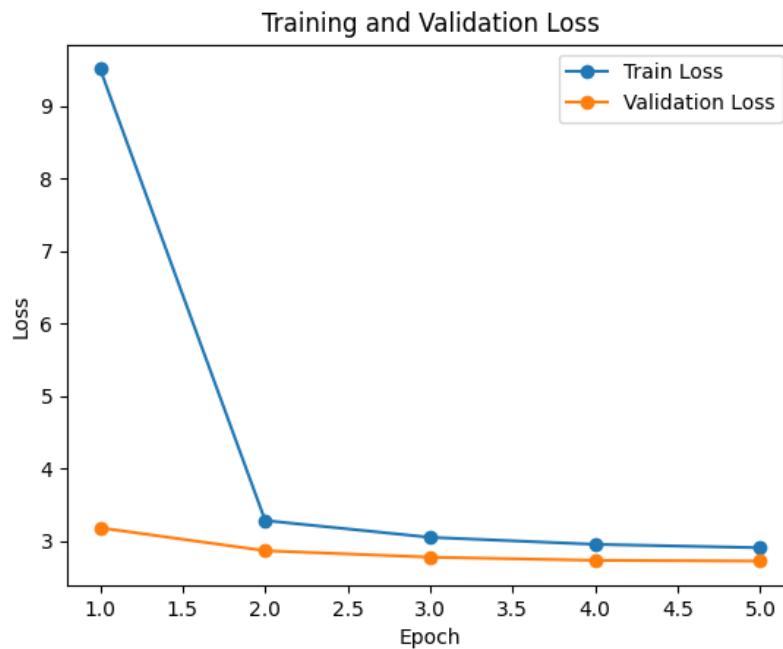


Figure 1: Training and validation loss versus epoch

The graph shows that the model is learning normally during training. Both the training loss and validation loss decrease over time, which means the model is improving its predictions as training continues.

The training loss started at approximately 9.5 during epoch 1 and decreased to around 2.9 by epoch 5. The validation loss also decreased from about 3.2 at epoch 1 to approximately 2.7 at epoch 5. Since both curves move downward together, the model appears to be learning useful patterns from the TinyShakespeare dataset.

There is no strong evidence of overfitting because the validation loss does not begin increasing at later epochs. Instead, it stays close to the training loss throughout the training process. If the model were overfitting, the training loss would continue decreasing while the validation loss would start increasing.

The biggest improvement happens between epochs 1 and 2, where the training loss drops sharply. After that, the decrease becomes slower and more stable, which is expected as the model gradually fine-tunes its understanding of the text.

Overall, the training process appears stable and effective.

2 R.2 Generated Text Analysis

This section analyzes the generated text from checkpoints at epochs 1, 3, and 5 using temperature 1.0.

Epoch 1

At epoch 1, the generated text mostly contains random characters, incomplete words, and broken sentence structure.

Example excerpt:

“sle , I r minlsey ot f opt aaIs osiilager soweillr nth t...”

Some English-like patterns appear, but most of the text is still difficult to understand. The model has mainly learned very basic character relationships and common letter combinations.

The text does not yet contain proper sentences or meaningful grammar.

Epoch 3

By epoch 3, the generated text becomes more readable. The model starts producing recognizable English words and short phrases.

Example excerpt:

“I ance fo mar thinn oic bb k wM h’: W,, lar we rd...”

At this stage, the model begins learning:

- common English spelling patterns,
- spaces between words,
- punctuation placement,
- short word fragments,
- dialogue-like formatting.

Although the text is still not fully coherent, there is a noticeable improvement compared to epoch 1. Some words and sentence fragments begin to resemble natural English.

This is the point where the model starts generating recognizable English words more consistently.

Epoch 5

At epoch 5, the generated text becomes more structured and contains longer readable phrases.

Example excerpt:

“thisy fand fre t tther oangenost sat, ishangle I nd e...”

Another example:

“nowoavoo ddy CYrnoe ogarsn-noroleareit s be...”

The model now produces:

- longer word sequences,
- more realistic spacing,
- punctuation,
- partial sentence structure,
- Shakespeare-style formatting patterns.

Even though many words are still misspelled or meaningless, the overall structure is noticeably better than earlier checkpoints.

The model begins producing recognizable English words around epoch 3, while partial sentence structures start appearing by epoch 5.

Overall, the generated text clearly improves as training progresses.

3 R.3 Temperature Analysis

This section compares generated text from the epoch-5 checkpoint using temperatures 0.5, 1.0, and 1.5.

Temperature 0.5

At temperature 0.5, the generated text is more repetitive and conservative.

Example excerpt:

“the wither t t thar t t the t oundoro tou the we...”

The model strongly prefers high-probability characters, which makes the output more stable but less creative. Many repeated patterns appear, and the text tends to loop through familiar sequences.

Temperature 1.0

At temperature 1.0, the text becomes more balanced between coherence and diversity.

Example excerpt:

“thisy fand fre t tther oangenost sat, ishangle I nd e...”

The output contains more variation while still maintaining some readable structure. This temperature produces the best overall balance for this model.

Temperature 1.5

At temperature 1.5, the generated text becomes much more random.

Example excerpt:

“K uPNoond woWn YYil kooerrfo E:”

Another excerpt:

“LChec g ttffudrangeos Ityans.”

The text contains many unusual letter combinations and nonsensical words because lower-probability tokens are sampled more frequently.

According to the temperature scaling equation:

$$P_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

higher temperature values flatten the probability distribution, making unlikely characters more likely to be selected during sampling. This increases diversity and randomness but reduces coherence and readability.

Lower temperatures sharpen the distribution, causing the model to choose safer and more predictable outputs.

Overall, temperature 1.0 produces the best trade-off between readability and creativity.

4 R.4 Reflection

TinyGPT learns several important patterns about English text during training.

First, the model learns character-level statistics such as common letter combinations and spelling patterns. This allows it to gradually generate recognizable English words and spaces between words.

Second, the model learns punctuation and capitalization patterns from the Shakespeare dataset. It also learns dialogue formatting such as character names followed by speech lines.

Third, the model learns short-range sentence structure and local grammatical relationships. By later epochs, it can generate text that partially resembles natural English sentences.

However, the model still has several limitations. Since it is a relatively small character-level transformer, it struggles with long-term consistency and deeper semantic understanding. Many generated words are still incomplete or meaningless, and the model often loses coherence over longer passages.

The limited embedding size and small number of transformer layers also reduce the model’s ability to learn more complex language patterns.

A larger transformer model could improve performance in several ways:

- more layers could learn deeper language structure,
- wider embeddings could store richer representations,
- larger context windows could improve long-range consistency,
- subword tokenization could help the model learn meaningful word units more efficiently.

Overall, this assignment demonstrates how transformer models gradually learn the structure of language through next-token prediction, even when trained on characters instead of words.